

Design and Implementation of a Compiler with Machine-Learning-Guided Optimization Selection

Kedar Sedai

Abstract—This paper documents the complete design, implementation, and evaluation of MiniLang, a research compiler for a minimal statically typed imperative language. MiniLang was build from scratch in C++17 to support end-to-end study of compiler construction and machine-learning-guided optimization. The system implements a classical multi-phase pipeline: hand-written lexical analysis, recursive-descent parsing, two-pass semantic analysis, lowering to a flat High-level Intermediate Representation (HIR), configurable rule-based optimizations, textual LLVM IR emission, and an ML-ready optimization advisor with evaluation harness.

We describe every compilation phase in implementation-level detail, including token design, abstract syntax tree (AST) structure, semantic rules, HIR instruction set, three optimization passes, LLVM code generation strategy, and a feature-driven advisor that selects optimization pipeline using heuristics, JSON plans or external Python interface. Experimental results in benchmark programs show that optimization reduces the HIR instruction count by up to 63% ($22 \rightarrow 8$ instruction) and LLVM IR size by 41% ($34 \rightarrow 20$ lines) on constant-heavy workloads, while preserving correct runtime behavior (verified by process exit codes.) Minilang is intended as a reproducible research and teaching traditional compiler engineering and modern ML-for-compilers research

Index Terms—Compilers, intermediate representation, LLVM, static analysis, code optimization, machine learning, pass ordering

I. INTRODUCTION

A. Motivation

Optimizing compilers transform source programs through dozens of analysis and transformation passes. Deciding which passes to run, and in what order has been tuned manually for decades in systems such as GCC and LLVM[1],[2]. Recently, machine learning (ML) has been applied to predict profitable compiler decisions—including pass ordering, heuristics flags, and inlining thresholds—from program features[3]–[6]. However, experimenting with ML-guided optimization typically requires navigating industrial-strength compiler infrastructures whose complexity obscures the relationship between front-end structure, IR design, and optimization outcomes.

Minilang addresses this by providing a complete but minimal compiler where every phase is inspectable, independently executable, and instrumented for measurement. This project follows a staged research methodology:

- Language design - define a small typed language sufficient for control flow, functions, and arithmetic.

- Front-end construction- implement lexer, parser, and semantic analyzer with explicit error reporting.
- IR design - lower AST to flat HIR with explicit temporaries and structured control flow.
- Rule-based optimization - implement classic local optimizations at HIR level.
- Code generation - emit human-readable LLVM IR for execution via Clang.
- ML advisor integration - extract static features, select pass pipelines, and evaluate against baselines.

B. Research Questions

This work investigates:

- RQ1: Can a minimal imperative language support a full, pedagogically transparent compilation pipeline comparable to textbook architecture [1],[7]?
- RQ2: Is a flat HIR with explicit temporaries a suitable layer for both rule-based optimization and feature extraction?
- RQ3: Can static HIR metrics drive selective application of optimization passes with measurable IR reduction?

C. Contributions

- Formal specification of the MiniLang language (lexical, syntactic, and semantic rules).
- Complete architectural documentation of a C++17 compiler with six standalone driver tools.
- HIR specification with lowering algorithms for expressions, conditionals, and loops.
- Three HIR optimizations with fixed-point iteration and statistics.
- ML-ready advisor framework: 14-dimensional feature vector, pass, registry, heuristic/JSON/Python modes, and minilang_eval harness.
- Empirical evaluation demonstrating IR reduction and correctness on benchmark programs.

D. Paper Organization

Section II reviews related work. Section III defines MiniLang. Section IV presents system architecture. Sections V–X detail each compilation phase. Section XI covers the ML advisor. Section XII presents evaluation. Section XIII–XIV discuss limitations and conclude.

II. RELATED WORK

A. Compiler Construction Foundations

The classic compiler pipeline- lexical analysis, syntax analysis, semantic analysis, intermediate code generation, optimization and code generation- is established in Aho et al.[1] and Cooper and Torczon [2]. MiniLang adopts this structure verbatim, using a hand-written scanner and recursive-descent parser rather than generator tools(Flex/Bison) to maximize transparency for readers and students.

Appel [8] and Muchnick [9] emphasize the role of intermediate representation in decoupling front-end language features from back-end code generation. Our HIR follows the principle: it is lower than AST(flat instruction list) but higher than LLVM (no SSA no target- specific details).

B. LLVM as a Back-end

Lattner and Adve introduced LLVM as a modular compilation framework with SSA-based intermediate representation and aggressive optimization infrastructure [2]. Rather than linking against the LLVM C++ API initially, MiniLang emits textual LLVM IR (.ll files), which Clang consumes for native code generation. This approach mirrors the LLVM Kaleidoscope tutorial [10] and keeps the research artifact readable in published papers.

C. Machine Learning for Compiler Optimization

Ashouri et al. provide a comprehensive survey of ML applied to compiler autotuning, pass ordering, and heuristic selection [3]. CompilerGym exposes LLVM optimization as a reinforcement-learning environment [4]. Other work learns graph representations of programs for heuristic tuning [5],[6]. MiniLang differs from these systems in scope and interpretability: optimization decisions occur at HIR level with explicit, low-dimensional features (14 integers), making it suitable for classroom demonstration and lightweight ML prototyping without requiring GPU-scale training infrastructure. The advisor is architected as a pluggable policy (heuristic today; ONNX or Python model tomorrow), aligning with survey recommendations for separating feature extraction from inference [3].

D. Educational Compilers

Educational compilers traditionally prioritize clarity over performance. MiniLang extends the educational pattern by adding (1) a documented semantic analyzer with symbol tables (2) a serialized HIR dump format, and (3) quantitative evaluation tooling- bridging teaching and research.

III. THE MINILANG PROGRAMMING LANGUAGE

MiniLang is intentionally small. It supports integer and boolean computation, functions conditionals, and loops. but omits pointers, arrays, heap allocation, and I/O builtins.

A. Lexical Structure

Category	Rules
Whitespace	Space, tab, newline; tracked for source locations
Comments	// to EOL; /* ... */ block comments
Identifiers	[A-Za-z_][A-Za-z0-9_]*
Integers	[0-9]+ (decimal, signed via unary -)
Keywords	int, bool, void, if, else, while, return, true, false
Operators	+ - * / %, comparisons, && ! =
Delimiters	() { } [] ; ,

TABLE I: MiniLang Lexical Rules

The lexer classifies keywords via `classify_keyword()` after scanning identifiers. Multi-character operators(==, !=, ≤, ≥, &&, ||) are recognized using one-character lookahead (`match()`).

B. Context-Free Grammar

MiniLang grammar is LL(1)-compatible:

program	::= function*
function	::= type IDENT "(" [parameters] ")" block
parameters	::= type IDENT ("," type IDENT)*
block	::= "{" statement* "}"
statement	::= block type IDENT "=" expression ";" "if" "(" expression ")" statement ["else" statement] "while" "(" expression ")" statement "return" [expression] ";" expression ";"
expression	::= logical_or
logical_or	::= logical_and (" " logical_and)*
logical_and	::= equality ("&&" equality)*
equality	::= comparison (("==" "!=") comparison) *
comparison	::= term (("<" "<=" ">" ">=") term)*
term	::= factor (("+" "-") factor)*
factor	::= unary (("*" "/" "%") unary)*

unary	::= ("!" "-") unary call
call	::= primary
primary	::= INT "true" "false" IDENT ["(" [arguments] ")" "(" expression ")"
arguments	::= expression ("," expression)*
type	::= "int" "bool" "void"

Operator precedence increases toward primary:logical OR (lowest) through multiplicative operators (highest). Unary operators bind tighter than binary operators.

C. Abstract Syntax Tree

The AST uses a tagged-union style with explicit Kind enumerations:

- Expressions: IntLiteral, BoolLiteral, Variable, Binary, Unary, Call.
- Statements: Block, VarDecl, If, While, Return, Expr
- Top level: (*Program* $\rightarrow$ *vector*) of FunctionDecl

Each node carries a SourceLocation (filename, line, column) for diagnostics. Types are represented by TypeKind: Int, Bool, Void.

Example. The factorial program:

```
int factorial(int n) {
    if (n <= 1) {
        return 1;
    }
    return n * factorial(n - 1);
}

int main() {
    int x = factorial(5);
    return x;
}
```

parses into two FunctionDecl nodes with nested IfStmt, ReturnStmt, CallExpr. and VarDeclStmt nodes.

D. Static Semantics

The SemanticAnalyzer performs:

Pass 1 - Function registration: Build global map function_: (*name* $\rightarrow$ *return_type*, *parameter_types*, *location*). Reject duplicate definitions.

Pass 2 - Per-function checking: Maintain stack of scope maps (scopes_) for block- enter/block-exit, Rules include:

Construct	Rule
Variable declaration	Initializer required; initializer type must match declared type
Variable reference	Must be declared in enclosing scope (inner shadows outer)
Arithmetic (+, -, *, /, %)	Operands: int; result: int
Comparison	Operands: int; result: bool
Logical(&&, , !)	Operands: bool, result: bool
if / while condition	Must be bool
Function call	Callee must exist; arity and argument types must match
return	Expression type must match current function return type

TABLE II: Rules:

Error handling: Panic-mode flag suppresses cascading errors after the first error in a function. Messages format as: semantic error at file:line:col:message.

Known limitation: Reassignment (*x* = *expr*; without type) is not supported-only type name = *expr*; declarations. This simplifies semantic analysis for the initial research prototype.

IV. SYSTEM ARCHITECTURE

A. Pipeline Overview

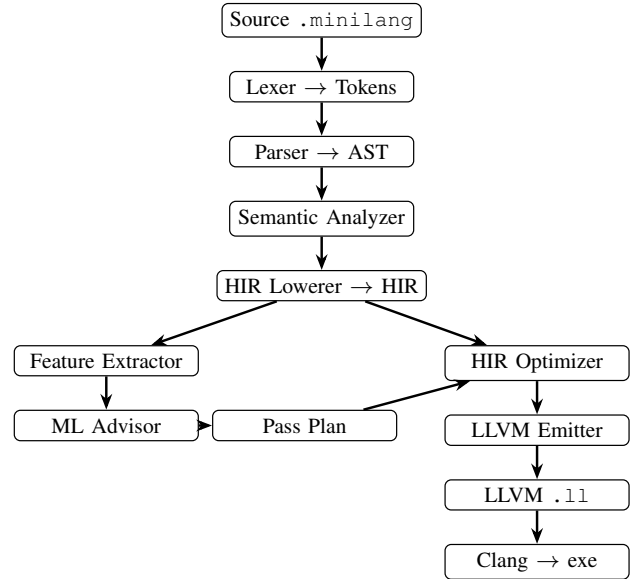


Fig. 1: Data flow.

B. Design Decisions

MiniLang is implemented in C++17 because it aligns with the LLVM ecosystem and is widely used in compiler courses and research. We use a hand-written lexer and recursive-descent parser instead of Flex/Bison so each compilation phase remains easy to read, debug, and explain in a research paper. The following table shows the key design choices.

Decision	Alternatives Considered	Rationale
C++17	Rust, Java	Native LLVM ecosystem; familiar in compiler courses [1]
Hand-written lexer/parser	Flex/Bison, ANTLR	Debuggability; direct mapping to textbook algorithms
HIR before LLVM	Direct AST → LLVM	Isolates front-end optimizations; simpler feature extraction
Textual LLVM IR	LLVM C++ API	Human-readable artifacts; faster research iteration
Separate drivers per phase	Single MiniLang CLI	Independent testing and paper screenshots per stage

TABLE III: Key Design Choices

V. PHASE 1: LEXICAL ANALYSIS

A. Scanner Architecture

The Lexer class maintains:

- `source_`, `filename_` - input buffer and path
- `current_`, `start_` - scan pointers
- `line_`, `column_` - position tracking
- `had_error` - error flag

Algorithm - Tokenization loop

- `skip_whitespace_and_comments()`
- If at end - return EOF token
- Record token start location
- If letter or '-' - `scan_identifier()` - keyword or Identifier
- If digit - `scan_number()` - `IntLiteral` with `int_value`
- Else switch on punctuation with lookahead for multi-chars
- On unknown char - Invalid token + error message

B. Comment and Whitespace Handling

`skip_whitespace_and_comments()` loops until non-skippable input:

- Whitespace: advance without emitting tokens
- `//`: consume until newline
- `/*`: consume until `*/` (non-nested)

C. Token Representation

Each Token contains:

- `TokenKind` (enum: `KwInt`, `Plus`, `EqEq`,...)
- `lexeme` (source spelling)
- `SourceLocation`
- `optional<int64_t> int_value` for integer literals

This design avoids re-parsing lexemes in the parser for numeric values.

D. Error Reporting

Invalid characters produce `TokenKind::Invalid` with descriptive messages. The lexer driver sets `had_error_`; compilation aborts with exit code 2 if lexer or parser errors occurred.

VI. PHASE 2: SYNTACTIC ANALYSIS

A. Parser Strategy

The Parser uses recursive descent: each grammar production maps to a `parse_*()` method. The parser holds `current_token` and provides:

- `match(kind)` - consume if current token matches.
- `consume(kind, message)` - require token or error
- `synchronize()` — Panic-mode recovery at statement boundaries (`;`, `}`, keywords)

B. Expression Parsing

Expression parsing follows the precedence chain: `parse_expression()` → `parse_logical_or()` → ... → `parse_primary()` Each level handles left-associative binary operators by looping:

```
// Conceptual structure for parse_term()
left = parse_factor();
while (match(Plus) || match(Minus)) {
    op = previous();
    right = parse_factor();
    left = make_binary(op, left, right);
}
return left;
```

Function calls use `finish_call()`: after parsing identifier `foo`, if (follows, parse argument list.

C. Statement and Function Parsing

- `parse_function()` - type, name, parameters, block body
- `parse_statement()` - dispatches on keyword or type lookahead
- `parse_var_decl()` - requires `=` and expression initializer
- `parse_if_statement()`, `parse_while_statement()` - requires parenthesized condition

D. Parser Output

Successful parse yields Program AST. `dump_program()` provides indented tree output for debugging and paper listings.

VII. PHASE 3: SEMANTIC ANALYSIS

A. Symbol Tables

Two symbol table structure:

- **Function table** (`unordered_map<std::string, FunctionSymbol>`) — global, populated in the registration pass.
- **Variable scopes** (`vector<unordered_map<string, VariableSymbol>>`) - stack; `enter_scope()` / `exit_scope()` on blocks and function entry.

Lookup walks scopes innermost-first, supporting shadowing.

B. Type Checking Algorithm

check_expression() returns TypeKind recursively.

AST node	Action
IntLiteral	Return Int
BoolLiteral	Return Bool
Variable	Lookup; error if undefined
Binary	Check operands per operator class; return result type
Unary	- requires int; ! requires bool
Call	Resolve function; check each argument type

TABLE IV: Table

check_statement() validates control-flow conditions and return types against current_return_type_.

C. Validation Tests

Negative test programs verify error detection:

- semantic_bad_undefined.minilang - use of undeclared variable.
- semantic_bad_type.minilang - type mismatch in expression

These are essential for demonstrating semantic soundness in the evaluation section.

VIII. PHASE 4: HIR LOWERING

A. Rationale for HIR

The AST is tree-structured and implicit about evaluation order and storage. HIR makes order explicit:

- Every subexpression result is stored in a numbered temporary (%t0, %t1, ...)
- Locals are named and accessed via LoadLocal / StoreLocal.
- Control flow uses explicit labels and branches.

This flat from simplifies data-flow analysis optimization [1, Ch. 9].

B. HIR Instruction Set

Opcode	Fields	Semantics
ConstInt	dest, int_value ;	dest $\leftarrow$ integer constant
ConstBool	dest, bool_value	dest $\leftarrow$ boolean constant
LoadLocal	dest, local_name	dest $\leftarrow$ memory[local]
StoreLocal	local_name, lhs	memory[local] $\leftarrow$ temp(lhs)
Unary	dest, lhs, unary_op	dest $\leftarrow$ op(temp(lhs))
Binary	dest, lhs, rhs, binary_op	dest $\leftarrow$ temp(lhs) op temp(rhs)
Call	dest, callee, args[]	dest $\leftarrow$ callee(args)
Ret	lhs	return temp(lhs) or void
BrCond	lhs, then_label, else_label	branch on temp(lhs)
Jump	jump_label	unconditional goto
Label	jump_label	branch target

TABLE V: HIR Instruction Set

Each HirFunction store: name, return type, parameters, locals, list, local_types map, and ordered instruction vector.

C. Lowering Algorithms

Expressions: lower_expression() allocates dest = fresh_temp(), emits one defining instruction, returns dest temp id.

Variable declarations: Evaluate initializer -> StoreLocal.

If-else: Classic structured lowering:

- Evaluate condition -> cond_temp
- BrCond, cond_temp, then_label, else_label
- Emit then_label: lower then-branch; Jump endif
- Emit else_label (if present): lower else-branch
- Emit endif_label

While loops:

- Jump while_cond
- while_cond: evaluate condition; BrCond, body, end
- while_body: lower body; Jump while_cond
- while_end:

This produces back-edges detectable by the feature extractor as loop_hint_count.

D. Example HIR Fragement

For int x = 2 + 3; inside a function, lowering produces approximately:

```
%t0 = const_int 2
%t1 = const_int 3
%t2 = add %t0, %t3
store_local x <- %t2
```

After constant folding, %t2 may collapse to const_int 5.

IX. PHASE 5: HIR OPTIMIZATION

A. Pass Pipeline Architecture

HirOptimizer accepts a PassPlan:

```
struct PassPlan {
    vector<OptPassKind> passes; // execution
    order
    int max_iterations = 3; // fixed-point
    bound
};
```

Registered Passes:

- ConstantFold
- AlgebraicSimplify
- DeadTempRemove

The optimizer iterates up to max_iterations over all functions until no pass reports changes-a standard fixed-point scheme [2, Ch.10].

B. Constant Folding

For each Binary or Unary instruction, the optimizer locates defining instructions for operates via find_def(temp, func, before_index) . If operands are ConstInt or ConstBool, the operation is evaluated at compile time and replaced with a constant instruction.

Safety: Division and modulo by zero are not folded (avoid compile-time crash).

Statistics: Each successful fold increments stats.constant_folds.

C. Algebraic Simplification

Currently implements identify $x * \rightarrow 0$ (and semantic $0 * x$) regardless of whether x is constant. This pairs with constant folding on expressions like $1 + 1 * 0$.

D. Dead Temporary Removal

Algorithm:

- Build used set: scan all instructions for reference in lhs, rhs, args,
- Filter instructions: drop pure computations whose dest is not in used
- Preserve control-flow of dead code elimination of temporaries [1, Ch. 9].

E. Optimization Statistics

OptimizationStats tracks folds, algebraic simplifications, and dead temps removed-reported by minilang_hir and minilang_compile drivers for reproducible experiments.

X. PHASE 6: LLVM IR CODE GENERATION

A. *L^AT_EX*-Specific Advice

Please use “soft” (e.g., `\eqref{Eq}`) cross references instead of “hard” references (e.g., (1)). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don’t use the `{eqnarray}` equation environment. Use `{align}` or `{IEEEeqnarray}` instead. The `{eqnarray}` environment leaves unsightly spaces around relation symbols.

Please note that the `{subequations}` environment in *L^AT_EX* will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you’ve discovered a new method of counting.

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L^AT_EX can’t read your mind. If you assign the same label to a subsection and a table, you might find that Table I has been cross referenced as Table IV-B3.

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Do not use `\nonumber` inside the `{array}` environment. It will not stop equation numbers inside `{array}` (there won’t be any anyway) and it might stop a wanted equation number in the surrounding equation.

B. Some Common Mistakes

- The word “data” is plural, not singular.
- The subscript for the permeability of vacuum μ_0 , and other common scientific constants, is zero with subscript formatting, not a lowercase letter “o”.
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an “inset”, not an “insert”. The word alternatively is preferred to the word “alternately” (unless you really mean something that alternates).
- Do not use the word “essentially” to mean “approximately” or “effectively”.
- In your paper title, if the words “that uses” can accurately replace the word “using”, capitalize the “u”; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones “affect” and “effect”, “complement” and “compliment”, “discreet” and “discrete”, “principal” and “principle”.
- Do not confuse “imply” and “infer”.
- The prefix “non” is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the “et” in the Latin abbreviation “et al.”.
- The abbreviation “i.e.” means “that is”, and the abbreviation “e.g.” means “for example”.

An excellent style manual for science writers is [7].

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D. Identify the Headings

Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is “Heading 5”. Use “figure caption” for your Figure captions, and “table head” for your table title. Run-in heads, such as “Abstract”, will require you

to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

E. Figures and Tables

a) *Positioning Figures and Tables:* Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. 2”, even at the beginning of a sentence.

TABLE VI: Table Type Styles

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
copy	More table copy ^a		

^aSample of a Table footnote.



Fig. 2: Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

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Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2]. Refer simply to the reference number, as in [3]—do not use “Ref. [3]” or “reference [3]” except at the beginning of a sentence: “Reference [3] was the first ...”

Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

Unless there are six authors or more give all authors’ names; do not use “et al.”. Papers that have not been published, even if they have been submitted for publication, should be cited as “unpublished” [4]. Papers that have been accepted for publication should be cited as “in press” [5]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [6].

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